

This document gives a brief description of the steps involved in the creation of the datafiles provided.

Downloading:

There are two datasets to download

Soil Survey Data

<https://www.mass.gov/info-details/massgis-data-soils-ssurgo-certified-nrcs>

- Click the 'Shapefiles' button to download the full set of shapefiles, navigate to the '**SOILS_MUPOLYGON_TOP20.shp**' shapefile
- Import this dataset into QGIS

Parcel Data

<https://www.mass.gov/info-details/massgis-data-property-tax-parcels>

- Click 'Download Statewide data' and put in a request to download
- Once given access, find the file '**L3_TAXPAR_POLY_ASSESS_EAST.shp**'
- Import this dataset into QGIS

Information on the creation of all of the variables in these datasets is included on the hosting sites.

Intersection

- To compute the area of each intersection I first computed their areas individually, and then used QGIS' intersection tool, overlapping Parcels with Soil Survey areas. I then calculated the area of each of these new intersected sections, to allow for the computation of the share of a parcel covered by a soil area.
- Once this was complete the new intersection data was exported as a csv to R, as well as a fresh copy of the Parcel data along with its WKT codes to allow us to remap the data.

R Work

Refer to '**Soil_Parcel_Script.Rmd**' for the code used.

After loading the data I created several new variables:

- Firstly, '**PARCEL_SHARE**', which simply measures the share of a parcel being covered by a soil survey area
- Next, '**SLOPE_AVG**', which uses the midpoints of the values in each '**SLOPE**' category as an estimate of the group's average to allow for weighted calculations.
- Third, I created '**SOIL_RES_DEPTH**' which fixes an issue with the labelling of the '**CM2RESLYR**' variable, which measures depth to a soil restrictive layer.
 - This variable originally had an issue where there were an extremely high amount of 0s in the data. After conferring with the lead soil scientist on the data we learned that many of these 0s are actually representative of soil areas with no clear soil-restrictive layer for the first 200 cm. Thus this variable '**SOIL_RES_DEPTH**' was created to rename these locations to 200cm to differentiate from real 0s,

- I then created '**ABOVE_15**', a variable that shows the areas of land that have a slope greater than 15% using the alternative slope variable, '**SLOPE_1**'

Next, due to a lack of unique identifiers within the data I created my own by appending together a number of different columns from the parcel data into one string, called 'unique_id', this would allow the parcels to be grouped together later to allow us to perform operations on each separately. This was accomplished by looping through a process of adding values to the unique identifier and examining the number of duplicates remaining until only the pure duplicates were left. After removing the remaining duplicates I moved on to calculations, grouping by this unique id each time.

- '**testfor100**' - I first created a variable that calculates the total % area of a parcel covered by soil survey areas by summing their **PARCEL_SHARE** columns.
- '**slope variables**' - There are three variables on various slope factors. Firstly, '**slope1**' which measures the weighted average of the slope categories that each area of the parcel belongs to, using the '**SLOPE_AVG**' variable created earlier. '**slope2**' then measures the weighted average slope based on the numerical values provided directly in '**SLOPE_1**'. Finally, '**slope3**' measures the portion of a parcel with a slope above 15% as calculated using '**SLOPE_1**'.
- '**soil_restrict**' - I then calculated the average depth to a restrictive layer on a parcel using the '**SOIL_RES_DEPTH**' variable
- '**concentrations**' - Lastly, I calculated the average concentration of sand ('**sandyness**') and clay ('**clayness**') in a parcel using the '**SAND**' and '**CLAY**' variables provided.

Final Dataset

To create the final dataset, I first joined the calculation datasets together. I then created a fresh copy of the parcel data with a reduced number of variables along with the '**WKT**' data for each parcel (which allows QGIS to plot the data) and joined this set with the calculation data.

Lastly, I created two variables:

- '**is_full**' - Which identifies parcels that have above a total of 90% of their area being covered by soil plots
- '**caution**' - Which indicates parcels that have NA for their total area covered
 - From a visual inspection, these appear to be parcels that are completely out of the bounds of any soil survey areas

I then saved this dataset as a csv

Into QGIS

To plot the data in QGIS I added the csv as a Delimited Text Layer with the following settings (Image Attached (1)):

- Encoding: **UTF-8**
- CRS: **EPSG:26986 - NAD83 / Massachusetts Mainland**
- Geometry Field: **WKT**

After loading the data in I saved the final dataset as a shapefile, (Image Attached (2)).

File name: F:\WorkStudyData\CH40b\Final\Soil_Parcel_Data.csv

Layer name: Soil_Parcel_Data Encoding: UTF-8

File Format

☐ CSV (comma separated values)
 ☐ Tab
 ☐ Colon
 ☐ Space

☐ Regular expression delimiter
 ☐ Semicolon
 ☒ Comma
 Others:

☒ Custom delimiters
 Quote:
 Escape:

Record and Fields Options

Geometry Definition

☐ Point coordinates Geometry field: WKT

☒ Well known text (WKT) Geometry type: Detect

☐ No geometry (attribute only table) Geometry CRS: EPSG:26986 - NAD83 / Massachusetts Mainland

Layer Settings

☐ Use spatial index
 ☐ Use subset index
 ☐ Watch file

Sample Data

	WKT
1	MULTIPOLYGON (((239852.893200003 912726.017999999,239854.821900003 912718.052200001,239854.857309427 912717.079267173,239854.73571042
2	MULTIPOLYGON (((216502.292400002 934439.192299999,216503.187700003 934436.068100002,216504.064000003 934432.867699999,216449.77139999

Close Add Help

(1)

Save Vector Layer as...

Format: ESRI Shapefile

File name: F:\WorkStudyData\CH40b\Final\Soil_Parcel_Data_Shape.shp

Layer name:

CRS: EPSG:26986 - NAD83 / Massachusetts Mainland

Encoding: UTF-8

☐ Save only selected features

Select fields to export and their export options

☒ Persist layer metadata

Geometry

Geometry type: Automatic

☐ Force multi-type
 ☐ Include z-dimension

☐ Extent (current: none)

Layer Options

RESIZE: NO

SHPT:

Custom Options

☒ Add saved file to map

OK Cancel Help

(2)